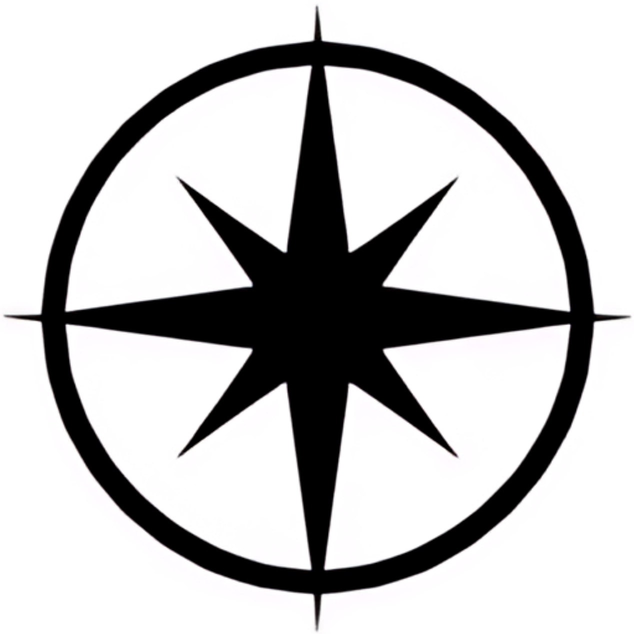

$$4^n - 1 = 3q$$

$$4^{k+1} - 1$$

$$4^k - 1 = 3q$$

$$4^k = 3q + 1$$

$$= 4 \cdot 4^k - 1$$

$$= 4(3q+1) - 1$$

$$= 12q + 4 - 1 = 12q + 3$$

$$= 3(q+1) = 3m \checkmark$$

$$n^2 + 3n = 2q \qquad k^2 + 3k = 2q$$

$$(k+1)^2 + 3(k+1)$$

$$= k^2 + 2k + 1 + 3k + 3$$

$$= k^2 + 3k + 2k + 1 + 3$$

$$= 2q + 2k + 4$$

$$= 2(q + k + 2) = 2m \checkmark$$

$$\frac{2a+3b}{5} \longrightarrow \frac{7a+8b}{5}$$

$$= (7a+8b)-(2a+3b)$$

$$= 7a-2a+8b-3b$$

$$= 5a+5b = 5q \checkmark$$

$$\frac{n^3+5n}{6} = k^3+5k=6q$$

$$(k+1)^3+5k+5$$

$$= k^3+k^2+2k^2+2k+k+1+5k+5$$

$$= (k^3+5k)+3k^2+3k+6$$

$$= 6q+3k^2+3k+6$$

note $3k^2+3k$ makes $3k(k+1)$ and $k(k+1)$ shows 2 integers one even and one odd multiplied resulting in an even integer and for that reason we can represent it as $2p$ hence $3(2p)$

$= 6q + 3(2p) + 6$

$= 6q + 6p + 6$

$= 6(q + p + 1) = \underline{6m}$ ✓